Higher-qubit QPT evidence audit

What survives, what must be relabeled, and what must be rerun

Date: 4 October 2026

Repository: `stochastic-frames` at `d62fcf192f27d8fb207ce61a1c466fc5059a7e1f`

> **Decision: the previous higher-qubit work is not worthless.** It remains valid evidence for a physically meaningful local-product-input, local-Pauli QPT benchmark. What does not survive is its attribution as a reproduction of Quiroga's reduced global-probe experiment, or the transfer of its tuned constants and performance claims to that different sensing geometry.

> **Main consequence.** The old engineering and scaling work is retained. The old empirical curves are retained and relabeled. Any headline claim about FRAMES versus Quiroga under Quiroga-style sensing must use the corrected campaign or be rerun. No historical artifact should be deleted.

1. Scope and audit method

This audit reviewed the active repository, Git history, documented sensing operators, tests, all named local QPT result trees, four shareable QPT PDFs, the verified September snapshot, and a bounded read-only inventory of the recorded Ruche result tree. No optimization was run and no historical artifact was modified or deleted.

- The active repository was clean and synchronized with `origin/stochastic-frames`.
- The legacy numerical implementation was checked against commit `423990f`; its core structured-data, operator, generator and GPU-runner files are unchanged at current `HEAD`.
- The Quiroga backend was added separately in `fe68eb4`, hardened in `74c968a`, and given a reproducible campaign runner in `d62fcf1`.
- Classification means: RETAIN as-is; RELABEL without discarding; RERUN before making the target claim; EXCLUDE from the primary comparison while preserving the record.

2. The two experiments are genuinely different

Feature	Previous higher-qubit benchmark	Current Quiroga-style benchmark
Input states	4^n product states from $\{\|0\rangle, \|1\rangle, \|+\rangle, \|+i\rangle\}^{\otimes n}$	$d^2=4^n$ global states: basis states plus all pairwise real and imaginary superpositions
Measurement	3^n local X/Y/Z settings with 2^n outcomes	One generalized 2d-outcome POVM
Rows	24^n; 331,776 at n=4 and 110,075,314,176 at n=8	$2d^3=2\cdot8^n$; 8,192 at n=4 and 33,554,432 at n=8
Coordinates	Normalized [I, X, −iY, Z] tensor basis	Choi/matrix-unit coordinates
Loss normalization	Half *mean* squared residual	Quiroga's printed rule uses half *sum* squared residual
Typical data audited	Fixed row-indexed additive Gaussian noise, commonly σ=0.05	Current n=2/n=4 campaign is noiseless
Physical scope	Conventional full QPT with product preparation and local readout	Reduced low-rank/near-unitary design; probes are already partly entangled at n=2

The objective-scale difference matters directly. The corrected campaign maps the mean-loss FRAMES convention to a fixed Quiroga TP weight using **$\beta=m/(2\lambda)$**. Old β, λ, step sizes, batch sizes and schedules therefore cannot be transferred numerically.

The geometry also changed. At n=4, the measured informative RMS scale of the reduced POVM is about 0.0023, versus a local-Pauli probability standard deviation around 0.058. The reduced design is much less well conditioned for this optimizer. Its failure is not explained by "fewer rows" alone.

Physical meaning

The old sensing matrix is not an arbitrary or unphysical random matrix. It comes from valid product input states and local Pauli outcome projectors. Its results remain physically interpretable as a simulated local-Pauli QPT protocol. The Gaussian observation model is synthetic and can yield values outside [0,1], so it is not literal finite-shot data.

The new design is mathematically physical—the POVM is positive and complete— but the repository models its probabilities rather than a device circuit. Preparing its global probes and implementing the generalized POVM would require nonlocal operations and possibly an ancilla dilation. Its 2d-outcome POVM is informationally complete for pure states, not generally for arbitrary high-rank mixed outputs.

> **Wording discipline:** the current n=4 campaign is a documented *Quiroga-style higher-dimensional extension*,
>
> not an exact published n=4 reproduction. Quiroga does not provide that experiment, the POVM constants, the
>
> adaptive-step proportionality constant, or the initialization normalization.

3. Campaign-by-campaign disposition

Campaign or evidence family	What was actually run	Disposition	Permitted claim
On-demand/scaling campaigns through n=10	Local-Pauli/QPT_BFW data, structured and product-state backends, synthetic noise, GPU profiling	RETAIN RELABEL	Valid memory, throughput, storage, backend-equivalence and local-Pauli empirical findings
n=8 fidelity-99 and coauthor package	One noisy local-Pauli channel/seed; verified factor fidelity 0.9900265	RELABEL	“99% factor-fidelity target reached on one noisy local-Pauli instance,” not exact recovery and not Quiroga sensing
Exponent comparison, n=8 scale tuning and 200k continuation	Controlled schedule/scale studies and exact continuation on the same local-Pauli family	RELABEL	Valid within-design optimization and engineering evidence; no cross-design hyperparameter transfer
n=8 FRAMES versus projected momentum-SGD	Complete held-out seeds 1–3 on paired local-Pauli minibatches	RELABEL RERUN if primary	Valid local-Pauli baseline. It cannot serve as the momentum baseline for the global-probe experiment without rerunning there.
n=8 MiFGD/Nesterov campaigns	Local-Pauli calibration; no scientifically promising admissible setting in the expanded penalty tuning	EXCLUDE primary	Retain as negative local-Pauli calibration evidence. Kim-style MiFGD is not the main Quiroga comparison.
n=8 “Quiroga adaFGD” campaign	Local-Pauli sensing plus a factor-action Frobenius proxy replacing the dense matrix spectral numerator; no admissible candidate	EXCLUDE primary	Algorithmic stress test only. It is neither Quiroga sensing nor the literal printed adaptive numerator.
n=4 “exact Quiroga” campaign	Literal dense adaptive numerator and corrected summed-loss mapping, but still evaluated on an 8,192-row subset of local-Pauli data	RETAIN audit EXCLUDE primary	The formula/finite-difference audit is valid. The empirical result is not a Quiroga-sensing result and is superseded by the current campaign.
Current n=2 and n=4 campaign at d62fcf1	Global probes, generalized 2d POVM, shared observations and initialization; printed adaFGD, FRAMES and Armijo controls	RETAIN primary	Current comparison evidence, with the published-constant and n=4 extension qualifications stated explicitly

Important recovered remote evidence

- The momentum campaign completed. FRAMES reached fidelities 0.99216, 0.98973 and 0.98919 on held-out local-Pauli sampling seeds; projected momentum-SGD remained near 8.73×10^{-6}.
- The first MiFGD grid enforced TP but remained near 7.6×10^{-6} fidelity; the coupled penalty retry had no candidate meeting its 0.1-fidelity gate.

- The n=8 proxy-adaFGD grid had no admissible candidate and retained all numerical failures.
- The n=4 legacy-sensing summed-loss retry had no scientifically promising candidate; its formulation audit nevertheless passed.

4. What the old “recovery” results mean under the current gate

The present noiseless gate requires fidelity ≥0.99, RMSE $\leq 10^{-6}$ and normalized TP $\leq 10^{-3}$, where normalized TP is $\| T(UU^\dagger) - I \|_F / \sqrt{d}$. The old noisy campaigns were not designed around this gate and generally did not record its RMSE. Applying it retrospectively is therefore a diagnostic, not a fair restatement of their original goal.

Legacy highlighted endpoint	Fidelity	Normalized TP	Current-gate interpretation
n=4 selected local-Pauli run	0.996561	0.000546	RMSE missing; cannot certify a full pass
n=5 selected local-Pauli run	0.906799	0.01369	Fails fidelity and TP
n=8 original “99%” endpoint	0.990027	0.01244	Real 99% factor overlap; TP is about 12.4× above the gate
n=8 requested-exponent continuation, 200k	0.991393	0.01066	Passes fidelity only; noisy local-Pauli result
n=10 at σ=0.05	0.0000401	0.3252	Fails
n=10 at reduced noise	0.990045	0.01595	Passes fidelity only and uses 32× lower noise

> **Interpretation:** no earlier higher-qubit endpoint is a documented pass of the present three-part gate. This

> does not erase the old n=8 result: it remains a verified high-factor-fidelity result on a noisy local-Pauli instance.

> It should be called that, rather than “exact recovery” or “Quiroga-style recovery.”

For comparison, the corrected n=4 held-out campaign also does not currently establish FRAMES recovery: FRAMES fidelity is 0.0129–0.0778 with normalized TP 0.00347–0.00354 at 2,000 effective row epochs. The printed adaFGD update passes TP but has fidelity 0.000789–0.01164. Exploratory 20,000-step Armijo FGD recovers all four seeds, demonstrating that the corrected inverse problem is solvable.

5. Existing reports and figures

Existing PDF	Action before reuse	What survives
`qpt_coauthor_report_20260915.pdf` 5 pages	Label every page “legacy local-Pauli/QPT_BFW benchmark.” Replace “recovery” headlines with factor-fidelity language unless the full gate is shown.	Numerics, provenance, local-Pauli curves and engineering discussion
`qpt_exponent_comparison_20260922.pdf` 11 pages	Relabel empirical pages 1 and 4–10. Do not transfer tuning conclusions to the global-probe design.	Historical schedules, within-design comparisons and provenance pages
`qpt_n8_scale_tuning_20260923.pdf` 9 pages	Relabel empirical pages 1 and 4–6.	Local-Pauli scale study; preparation-speed engineering with scoped wording
`qpt_n8_followups_20260923.pdf` 3 pages	Archive as a stale, unsubmitted local-Pauli proposal.	Historical planning/provenance only

Scientific plots requiring the same legacy label include `n8_fidelity_iterations`, `n8_sampling_and_tp`, `recovery.png`, the n=4/n=5 strong/weak figures, n=8 draw-budget figures, n=10, step-half, step-one and late-comparison plots. `preparation_speedup` may remain an implementation-engineering figure, but its measured speedup does not automatically apply to the new global-probe backend.

Manuscript impact

The repository contains no manuscript TeX or bibliography, so the actual Section 5.2 and manuscript-integrated figures could not be edited or checked. The required correction is already documented in `docs/qpt_quiroga_sensing.md`, lines 285–320.

Recommended manuscript wording.

> “We evaluate two distinct QPT sensing designs. Benchmark A uses product inputs and local Pauli measurements, giving 24^n probability rows; its measurement loss is mean-normalized. Benchmark B follows the reduced global-probe/2d-POVM construction of Quiroga and Kyrillidis, giving $2\cdot 8^n$ rows. The two designs have different signal scale and conditioning, so optimizer constants and row budgets are not transferred between them. Higher-qubit results from Benchmark A do not reproduce the experiments of Quiroga and

> Kyrillidis.”

6. Preservation completed during this audit

The existing preservation state was already strong:

- `/home/tony/qpt_sync_20260923_01` is a verified snapshot with 516 payload files and 321,831,297 payload bytes.
- Its archive SHA-256 is `9da5f968be405fbfc4ae94f62d81bc67051fa2b7023929941a4728eeffcacd2c`.
- All 480 active `results/qpt` files and all four QPT PDFs match the snapshot byte-for-byte.

Five later baseline campaigns existed completely on Ruche but only as launcher bundles on Elite. Because Ruche work storage is not backed up, this audit copied their complete stable result trees to:

`/home/tony/qpt_higher_qubit_audit_20261004_01/archives/ruche`

Preserved campaign	Files	Bytes	Full manifest digest
n8_momentum_sgd_20260924_01	76	53,635,205	`9c15f08f…aa2f`
n8_mifgd_20260925_01	19	10,713,588	`6e67ad0f…9821`
n8_mifgd_penalty_tuning_20260927_01	19	6,554,568	`0d5c0438…f1fe`
n8_quiroga_adafgd_20260927_01	87	14,272,419	`b15e6d83…e2b8`
n4_quiroga_exact_20260928_01	104	2,028,970	`b8e8a6bb…b862`

All 305 files and 87,204,750 bytes have identical per-file aggregate manifests on Ruche and Elite. The Ruche queue was empty before collection. Nothing was removed from Ruche. The complete machine-readable inventory is `artifact_manifest.json` beside this report.

7. Decision for the project

1. **Keep the old high-qubit work.** Present it as a separate local-Pauli/QPT_BFW benchmark and retain its engineering contributions.
2. **Correct attribution before reuse.** Do not call the old 24^n design Quiroga’s setup or say it faithfully reproduces that paper.
3. **Do not claim a previous strict-gate pass.** The older runs optimized noisy factor fidelity and approximate TP; they lack the present noiseless RMSE certificate.
4. **Use the d62fcf1 n=2/n=4 campaign as the primary current comparison.** State that n=4 is a principled extension, not a published reproduction.
5. **Rerun only claims that need the global design.** Momentum, MiFGD, higher-qubit FRAMES, noise robustness and sample-efficiency claims require new experiments if they are to appear in the primary Quiroga-style comparison.
6. **Do not jump directly to an expensive n=8 conclusion.** First determine at n=4 whether extended-budget FRAMES can converge and satisfy TP on the corrected geometry. That separates an iteration-budget problem from a structural fixed-penalty problem.

> **Final answer:** the earlier higher-qubit program survives as a substantial alternative physical benchmark

> and engineering result. What is currently unvalidated is the transfer of its performance story to Quiroga-style

> global sensing.

8. Limitations and evidence references

- No manuscript source was available in the repository.
- Old campaigns generally did not save the RMSE required by the new gate.
- Several highlighted old results use one channel/seed.
- The old Gaussian-noise and current noiseless campaigns answer different statistical questions.
- The current global-probe backend has no fused GPU runner, so legacy GPU scaling cannot be transferred.
- General higher-rank recovery is not certified by the 2d-outcome POVM.

Primary code/document evidence:
`docs/qpt_quiroga_sensing.md:15–31, 213–247, 274–320`
`paper/experiments/qpt_structured_data.py:58–75, 165`
`paper/experiments/qpt_quiroga_sensing.py:1–35, 148–255`
`paper/experiments/quantum_process_tomography.py:275`
`paper/experiments/qpt_quiroga_campaign.py:12, 481`
`tests/test_qpt_structured_data.py:90` and `tests/test_qpt_quiroga_sensing.py:83, 155, 161`

Audit status: complete. Repository unchanged and clean; historical artifacts retained; five later Ruche campaigns newly backed up and checksum-verified; no compute jobs submitted.